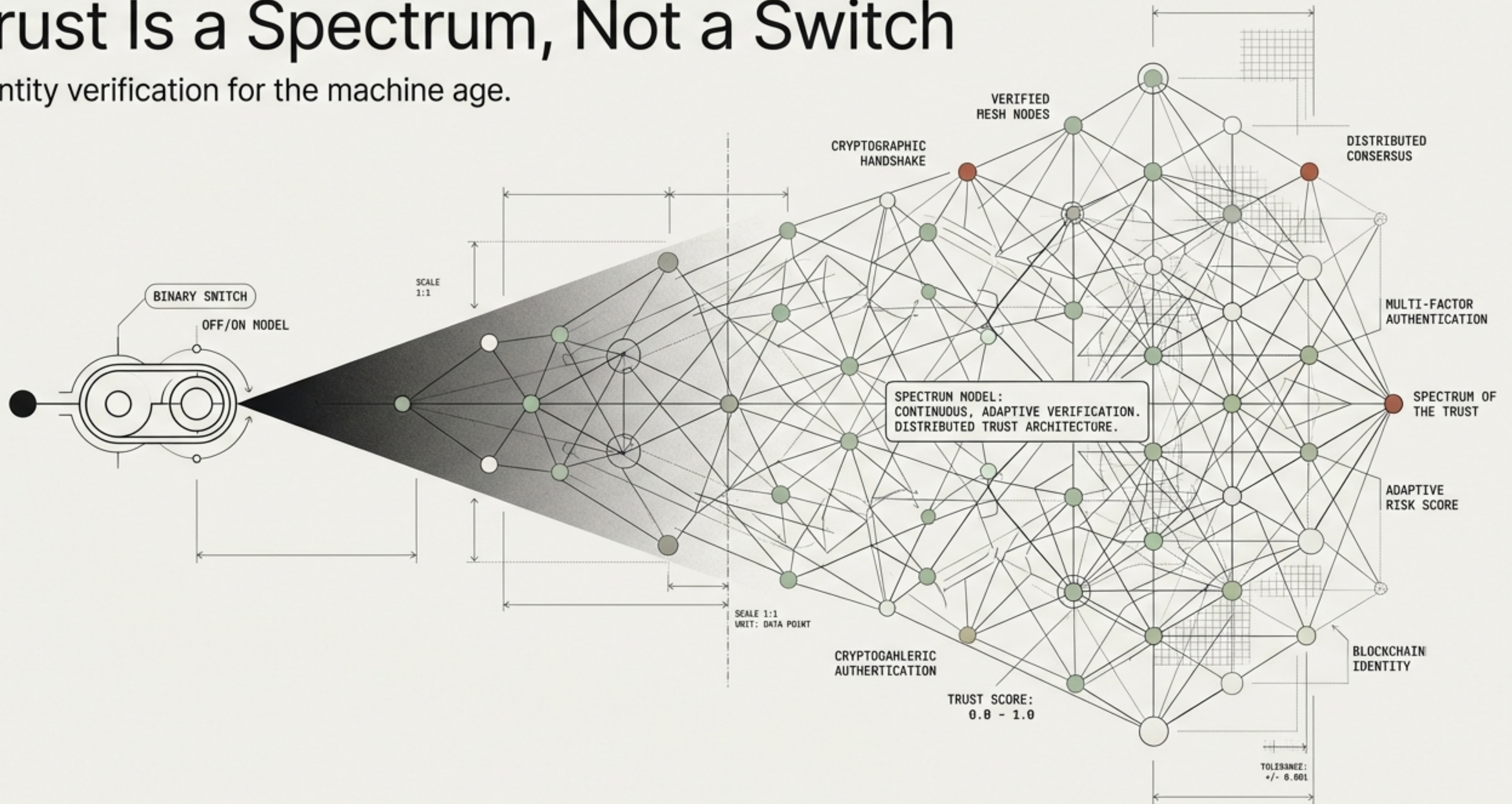


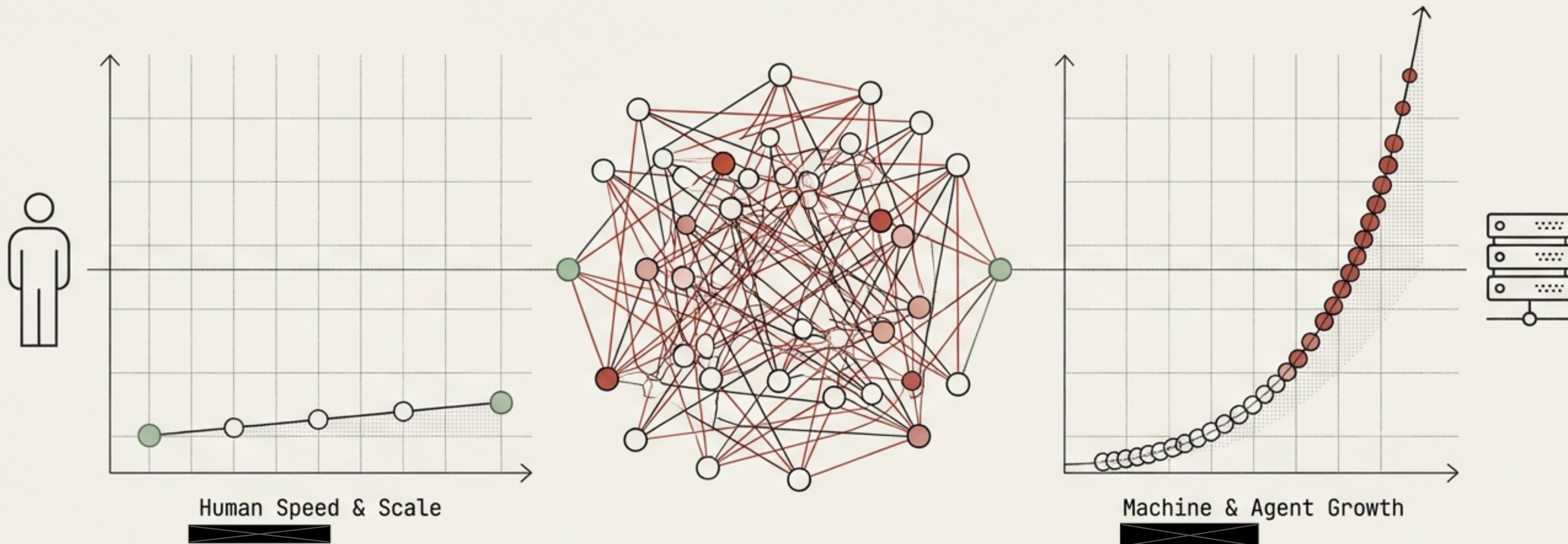
Trust Is a Spectrum, Not a Switch

Identity verification for the machine age.



Applying human metrics to a machine world creates a security catastrophe

The number of machines and AI agents is growing non-linearly. Requiring human verification at each intermediate step introduces impossible friction and systemic vulnerability.

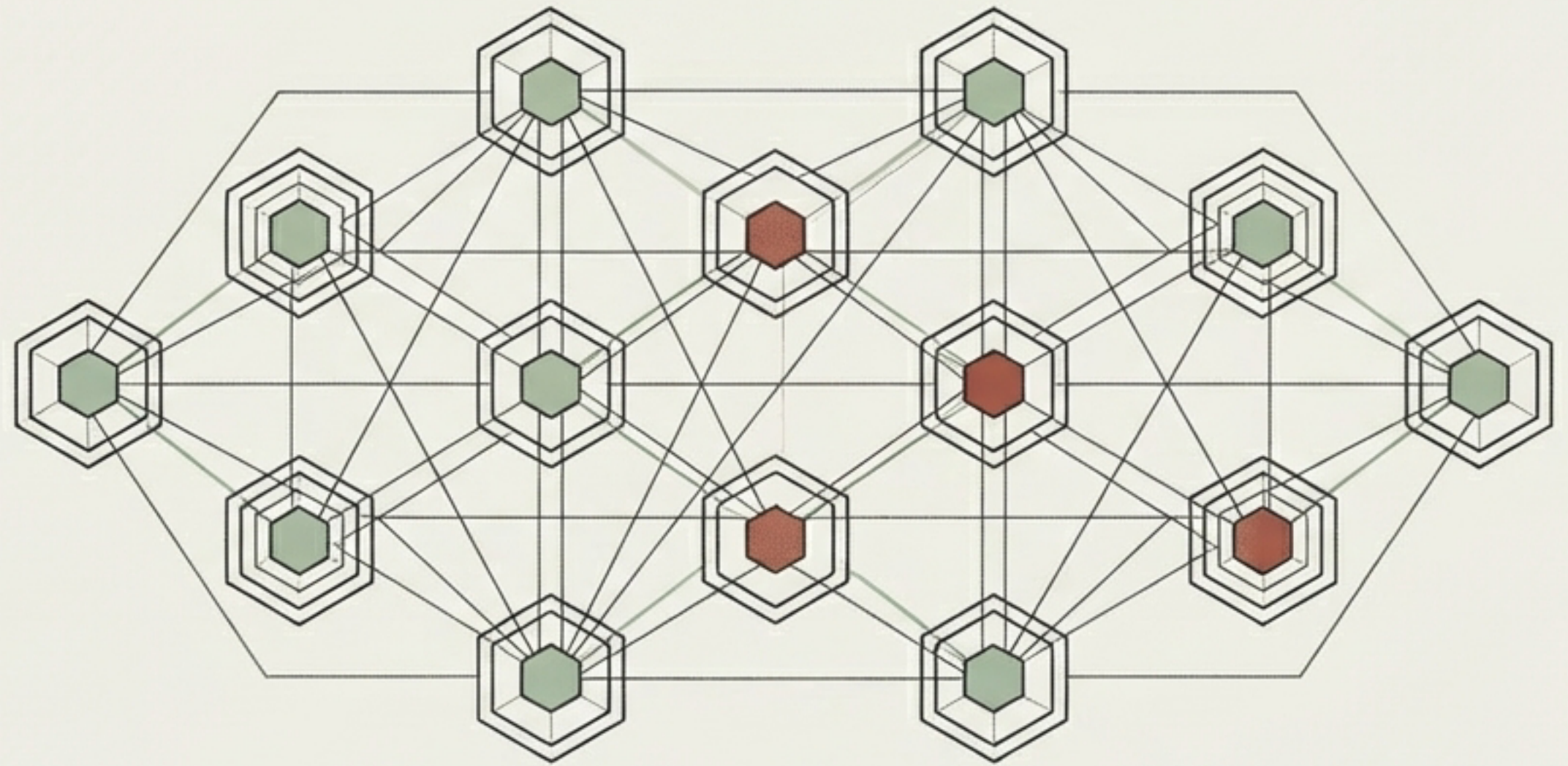


Human identity belongs at the edge



The Edge

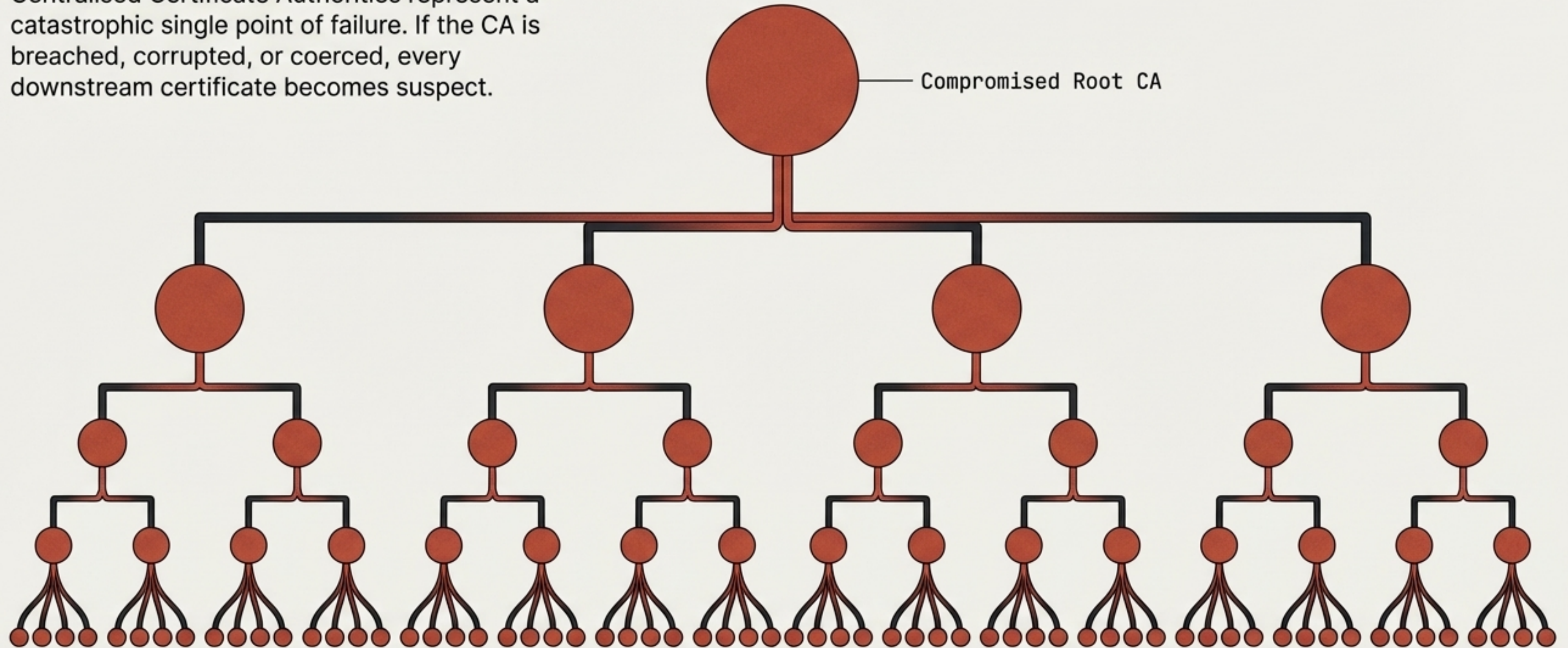
Human verification happens exactly once: at the moment a person asserts their presence to their own device. Everything past that boundary is pure machine identity and cryptographic delegation.



The Core

The instinct to centralise trust creates the exact vulnerabilities we want to avoid

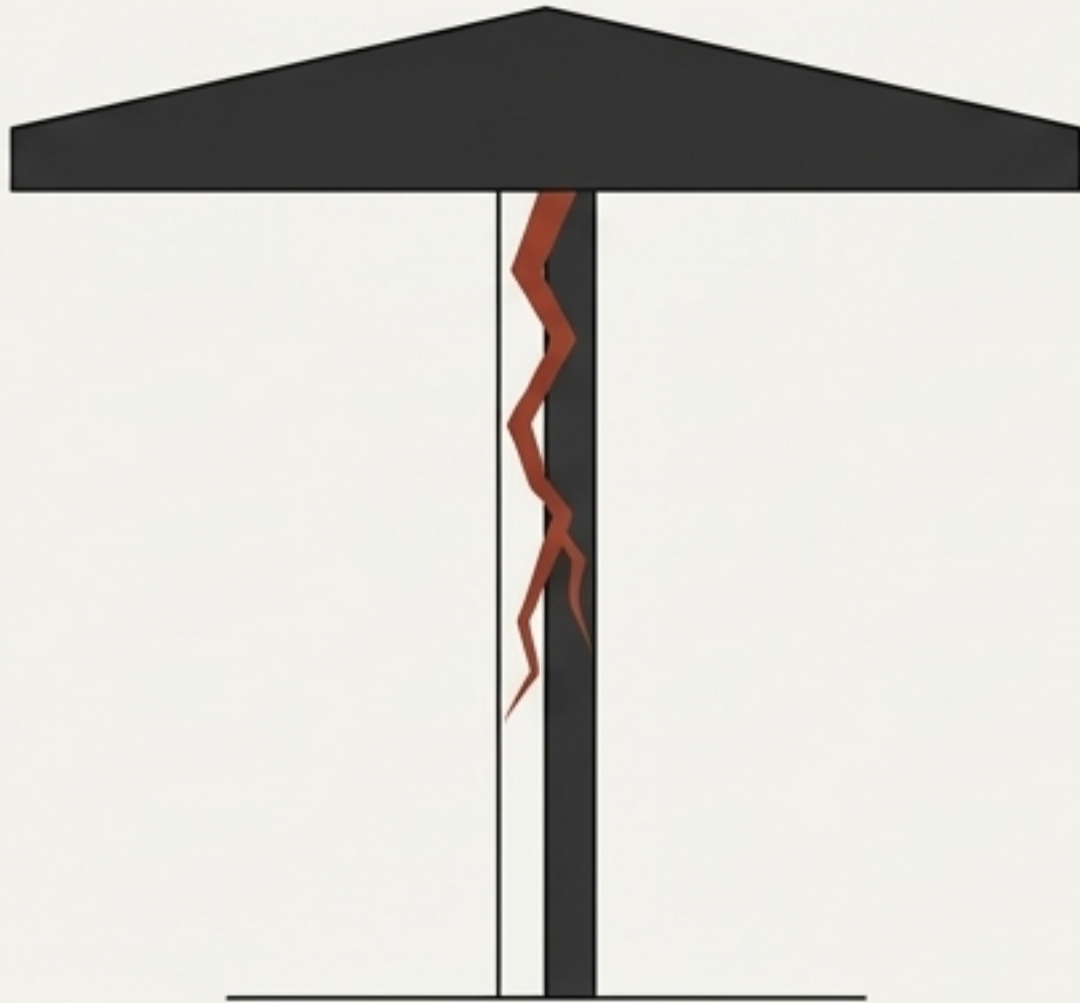
Centralised Certificate Authorities represent a catastrophic single point of failure. If the CA is breached, corrupted, or coerced, every downstream certificate becomes suspect.



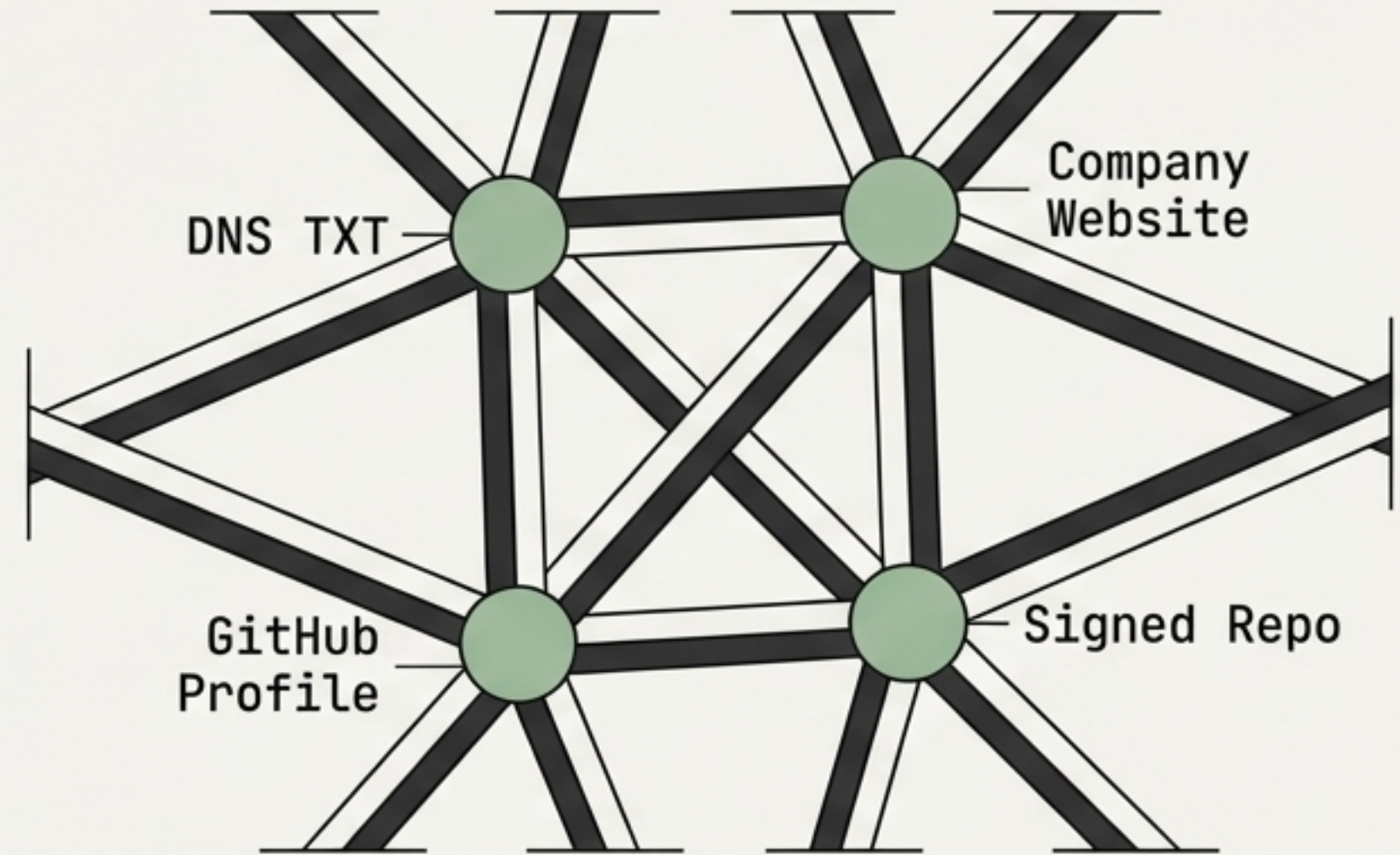
Trust does not require a central authority, it requires multiple independent witnesses

An attacker must simultaneously compromise your DNS registrar, web hosting, GitHub account, and repository. It is a fundamentally different threat model than compromising a single central root.

Hierarchy

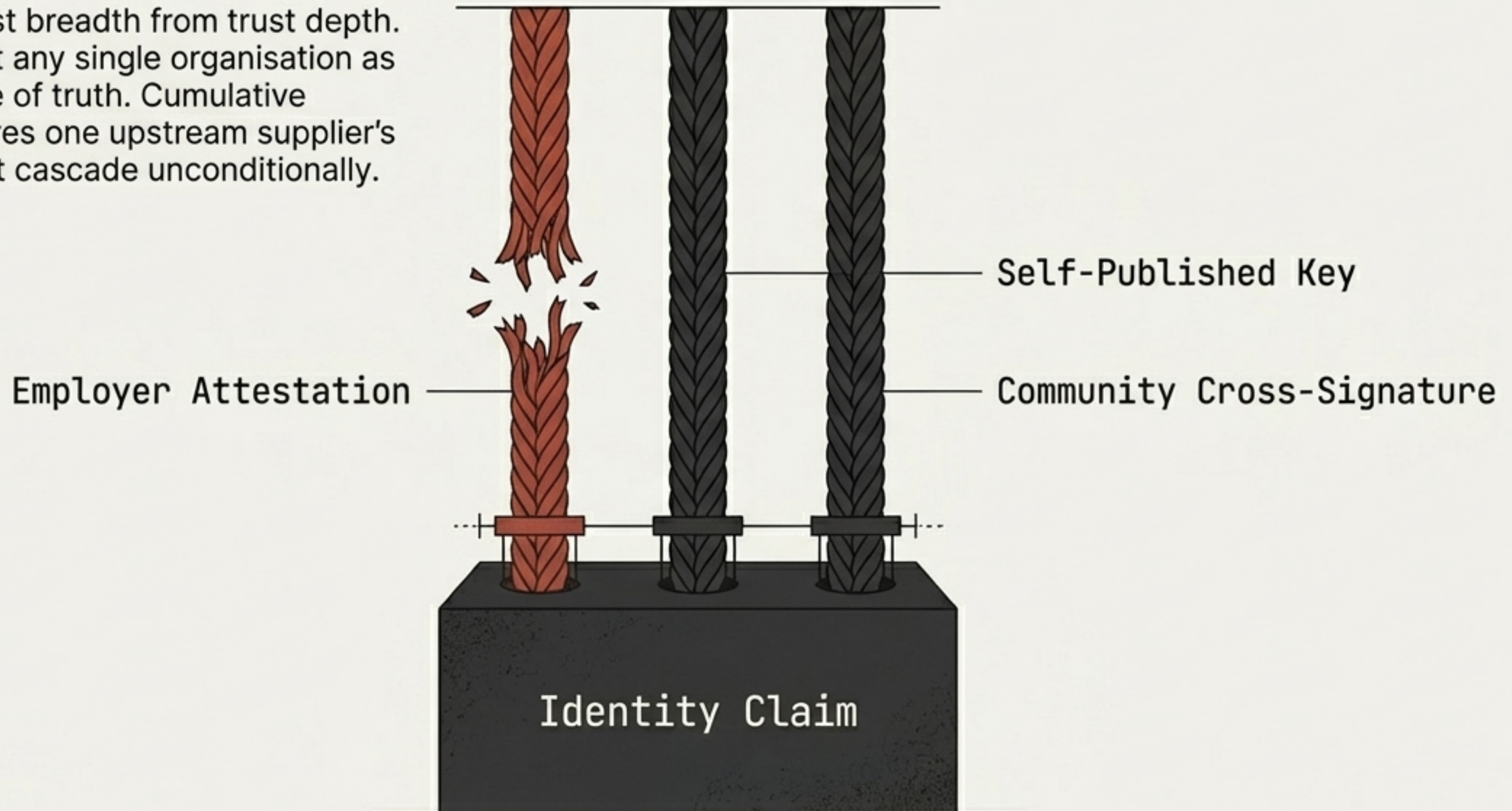


Mesh



The answer to sovereign risk is the elimination of single sovereignty entirely

Distinguish trust breadth from trust depth. We do not trust any single organisation as our only source of truth. Cumulative evidence ensures one upstream supplier's failure does not cascade unconditionally.



There is no fully trusted state

Trust is a continuous gradient based entirely on accumulated, independently verifiable evidence.



Is this trusted enough, for what we are about to do, given the evidence we have?

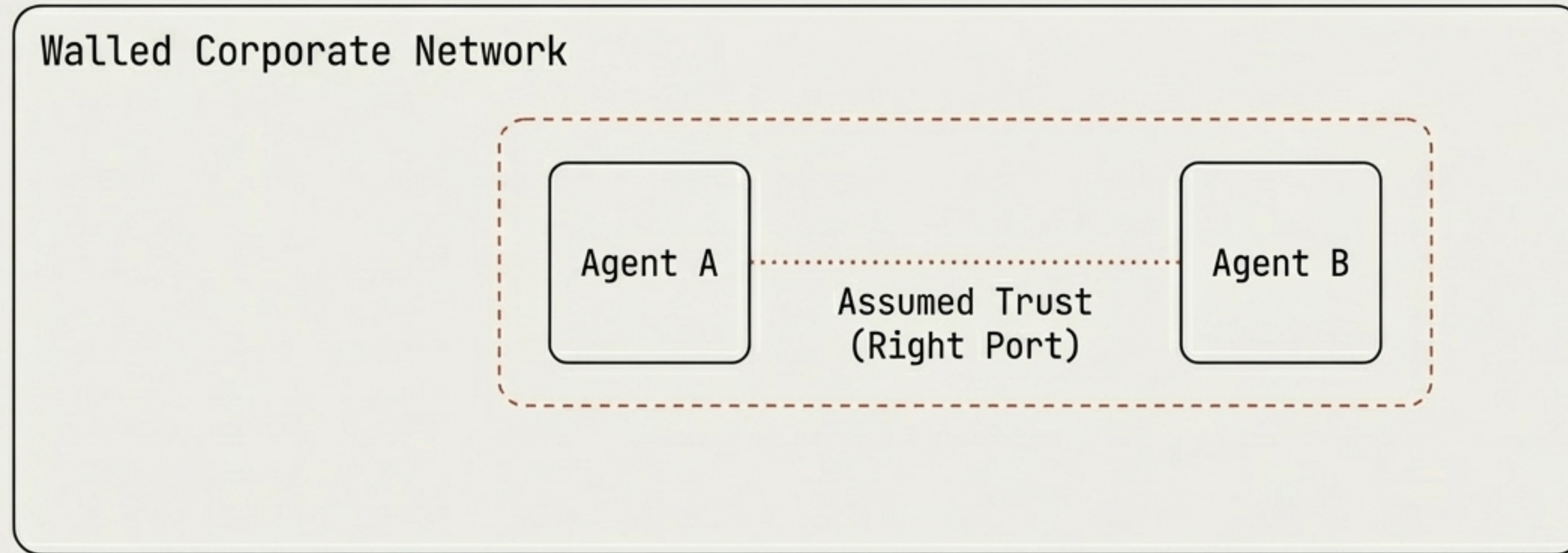
The required weight of evidence dynamically scales with the operational risk of the requested action.

	Low Evidence Sufficient to read an unknown LinkedIn message, but execution is blocked.
	Moderate Evidence Sufficient to open an attachment from a known colleague.
	High Evidence Sufficient to act on a request or elevate privileges following multi-channel verification.

AI agents currently authorise requests based on proximity, not proof

Agents are orchestrating tasks and interacting on our behalf today. Trusting internal systems simply because they exist on the same network is a naive model that inevitably fails at scale.

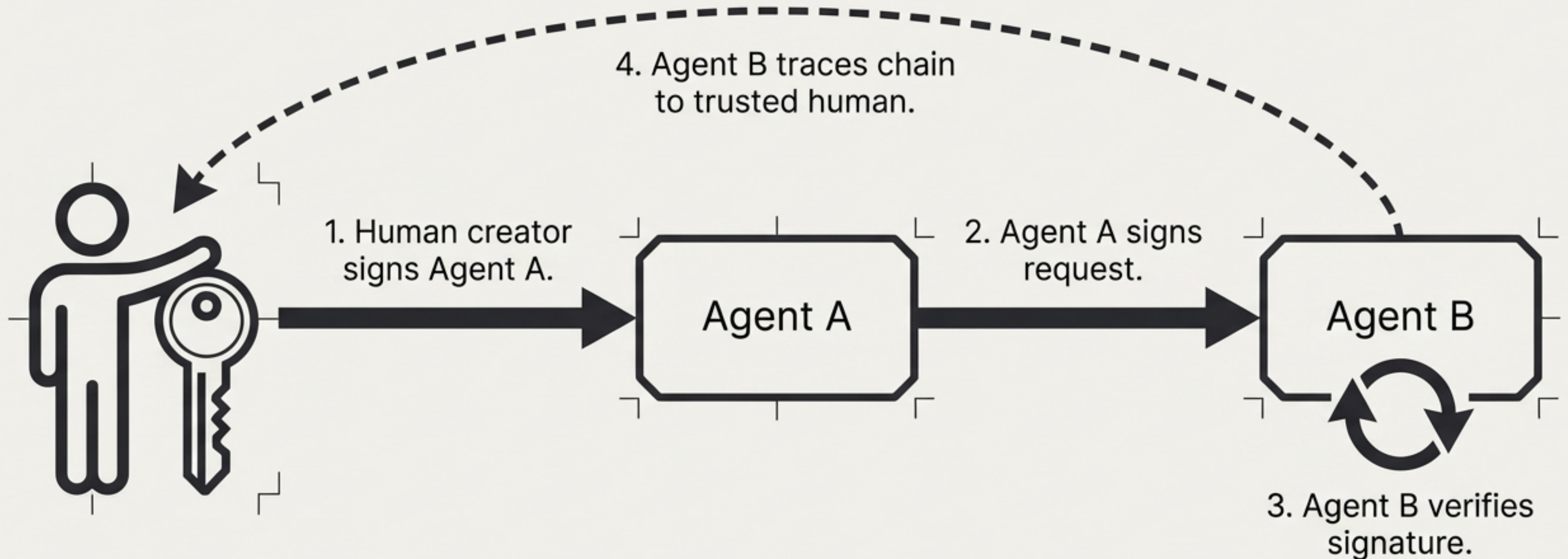
The Naive Trust Model



Mathematical trust flows without central authorities

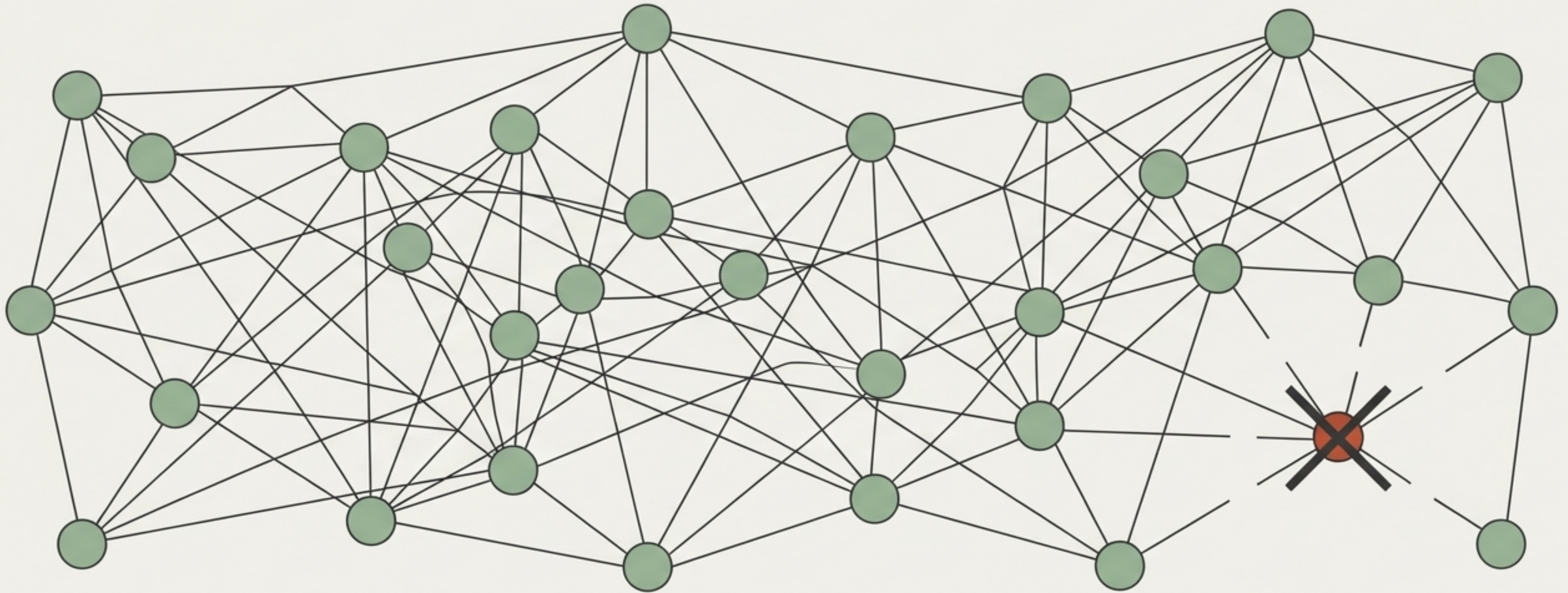
Each agent owns a key pair. Verification happens mathematically.

No agent needs to ping a central authority to validate the chain of trust.



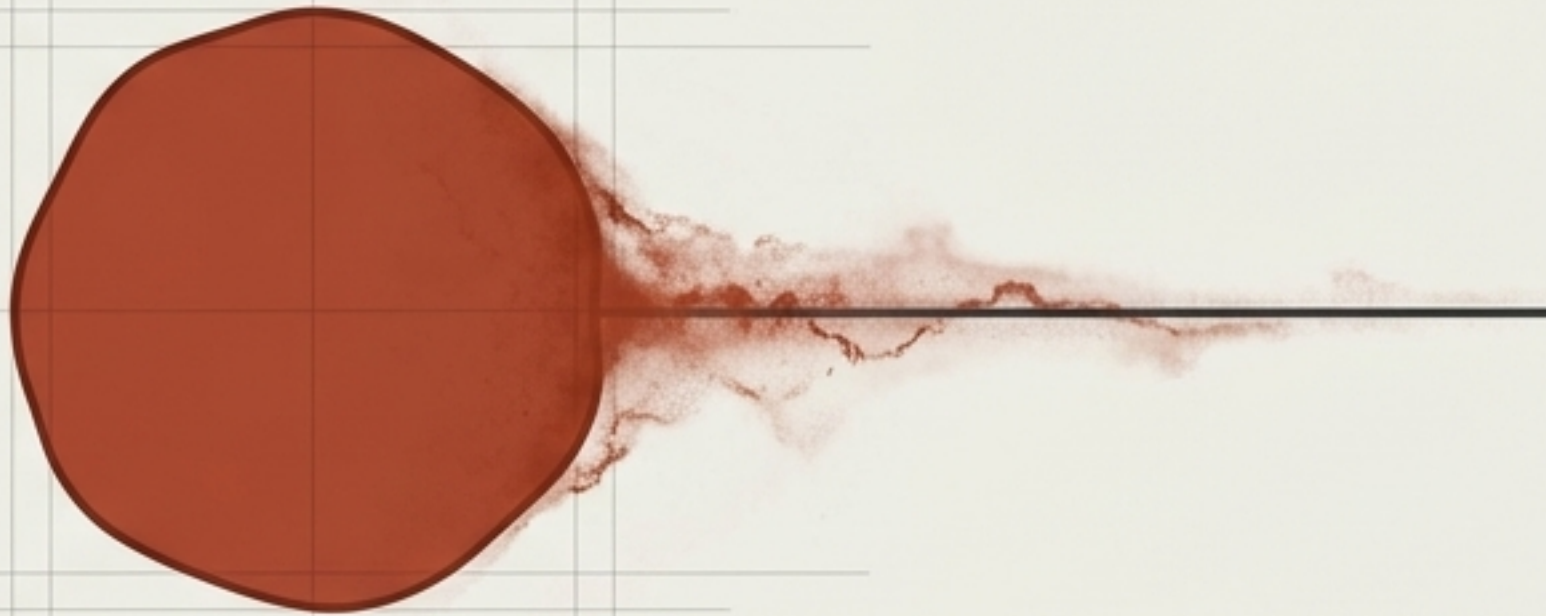
Containment without systemic collapse

Because keys are decentralised, there is **no central CA to compromise**. A compromised agent's key is revoked in isolation without affecting any other agent's identity.



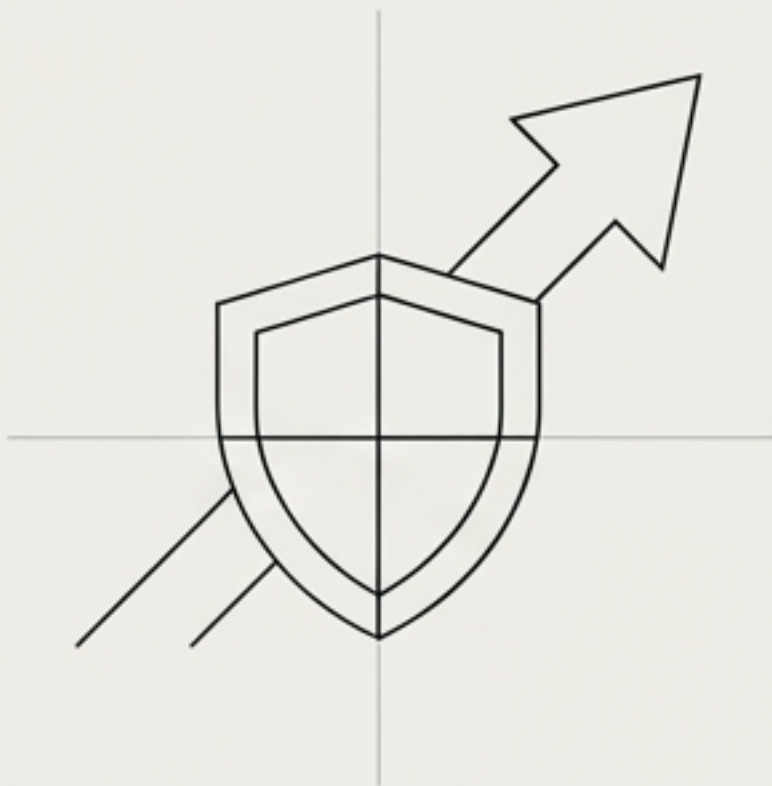
What happens when the root of trust is itself compromised?

If the initial human authentication is bypassed, the downstream chain starts from a poisoned root. Trust ultimately requires human judgment about the operators and creators of systems. That judgment is not mathematical, and it will sometimes be wrong.



Technology cannot eliminate human risk, but it makes it manageable

We build these systems not to achieve impossible perfection, but to aggressively shift the economic and structural advantage away from the attacker.



Raise the cost of attack



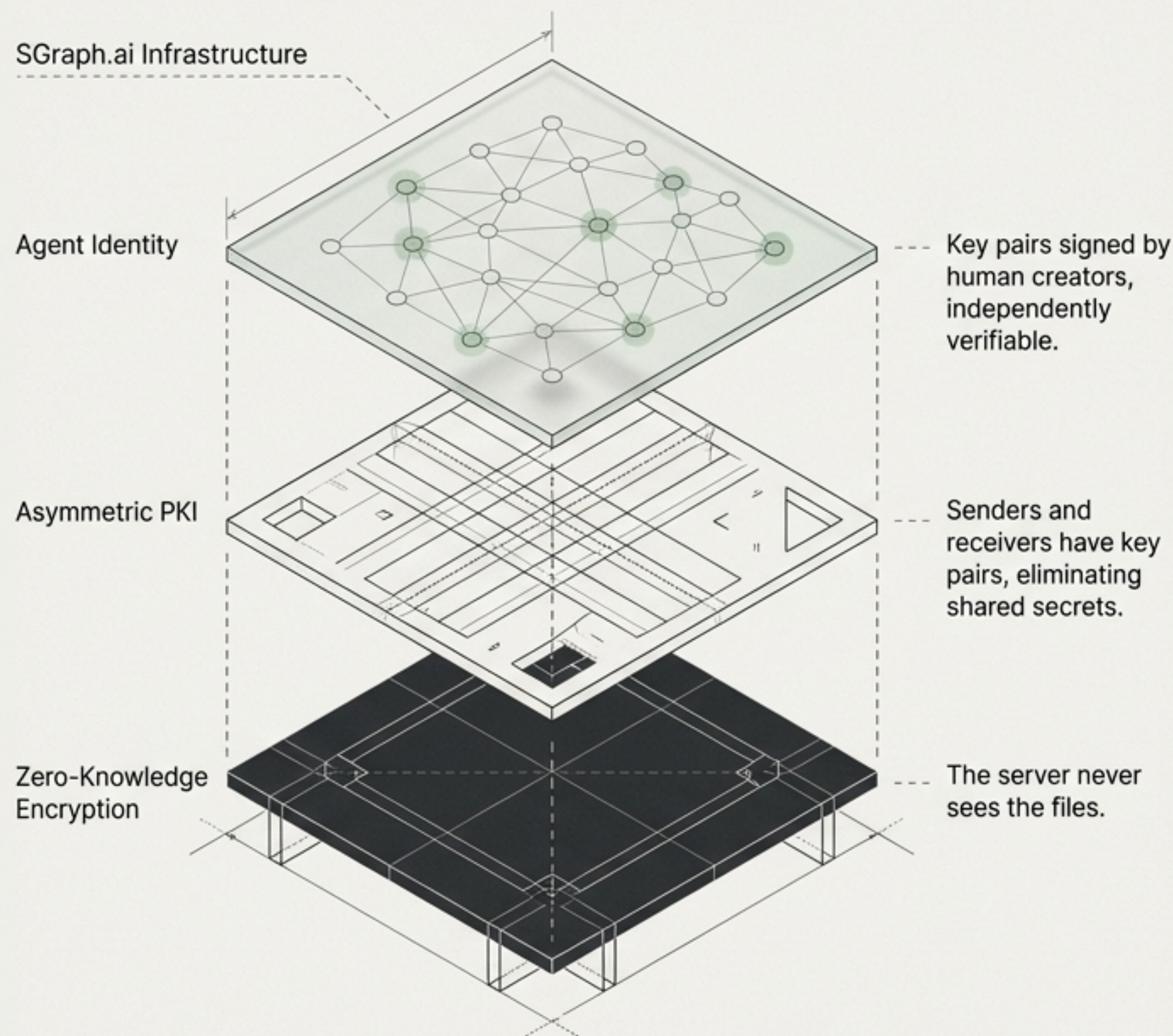
Distribute risk across independent nodes



Make the evidence for trust decisions visible

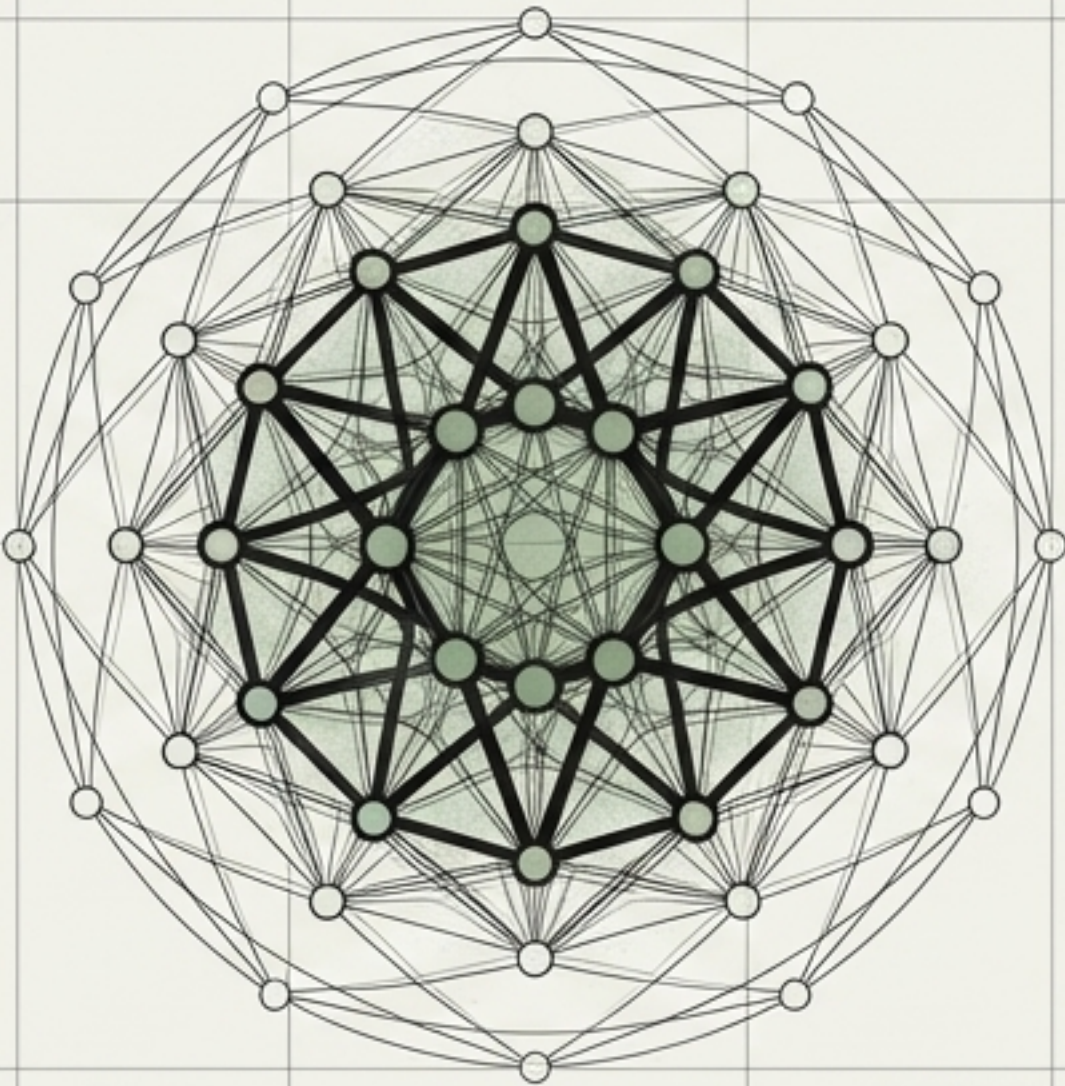
Verification built into the architecture, not bolted on afterwards.

We are building the infrastructure of trust in the open. Decentralised, auditable, and mathematically sound.



The web of trust is a practice, not a product

The system grows stronger every time a key is published, a message is signed, or a claim is independently verified.



Building in public at sgraph.ai

Code available at github.com/the-cyber-boardroom/SGraph-AI__App__Send
(Apache 2.0)